

PUPIL NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 3 • C1700–C1900

Medicine in 18th & 19th Century Britain

Jenner, Nightingale, Lister, Pasteur, Koch, and Snow: The Scientific Conquest of Disease



Primary Visual Source: *The Dawn of Bacteriology and Scientific Surgery* (c.1700–c.1900)

Accession Shelfmark: ARCH-MED-1885

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE	LEARNT	REVISED
Key Topic 3.1: Ideas on Causes: Pasteur's Germ Theory & Koch's Bacteriology <i>How did swan-neck flasks, agar jelly, and chemical dyes disprove spontaneous generation and reveal microbes?</i>	☐	☐
Key Topic 3.2: Approaches to Prevention: Edward Jenner & The Smallpox Vaccine <i>How did Jenner's 1796 cowpox experiment and parliamentary compulsion eradicate the greatest killer of children?</i>	☐	☐
Key Topic 3.3: Improvements in Hospital Care: Florence Nightingale & Professional Nursing <i>How did Scutari, Notes on Nursing (1859), and the Pavilion Plan transform hospital wards from death houses?</i>	☐	☐
Key Topic 3.4: The Surgical Revolution: Simpson's Chloroform & Lister's Antiseptics <i>How did Simpson conquer pain, Lister conquer infection, and the 1890s introduce aseptic surgery?</i>	☐	☐
Key Topic 3.5: Public Health & Cholera: John Snow & The 1875 Public Health Act <i>How did John Snow's Broad Street pump map (1854) and the 1875 Act permanently dismantle government laissez-faire?</i>	☐	☐

Living Unit Timeline: 18th & 19th Century Medicine (Part I: 1700–1854)

JENNER DEVELOPS SMALLPOX VACCINE

Edward Jenner inoculates James Phipps with cowpox pus from Sarah Nelmes, successfully protecting him from smallpox and founding immunology.

SIMPSON DISCOVERS CHLOROFORM

James Young Simpson discovers the potent anaesthetic properties of chloroform in Edinburgh, eliminating operative pain shock.

FIRST PUBLIC HEALTH ACT PASSED

Prompted by Edwin Chadwick's 1842 report on sanitary conditions, Parliament establishes a General Board of Health (permissive and optional).

JOHN SNOW & BROAD STREET PUMP

Dr John Snow uses his Soho cholera spot map to identify contaminated water from the Broad Street pump, disproving the miasma theory.

• LIVING TIMELINE SKETCHPAD & CAUSAL SYNTHESIS (1700–1854)

Sketch Jenner's cowpox experiment, Simpson's chloroform trials, or Snow's Broad Street pump map.

Living Unit Timeline: 18th & 19th Century Medicine (Part II: 1854-1900)

CHRONOLOGICAL FRAMEWORK • GERM THEORY, ANTISEPTICS & SANITATION

1854-56

NIGHTINGALE AT SCUTARI (CRIMEA)

Florence Nightingale enforces hospital sanitation, clean water, and fresh bedding, reducing British soldier mortality from 40% to 2%.

1861

PASTEUR PUBLISHES GERM THEORY

Louis Pasteur disproves spontaneous generation using swan-neck flasks, demonstrating that airborne microorganisms cause decay and disease.

1865

LISTER PIONEERS ANTISEPTIC SURGERY

Inspired by Pasteur's Germ Theory, Joseph Lister uses carbolic acid spray during operations, reducing amputee mortality from 46% to 15%.

1875

THE SECOND PUBLIC HEALTH ACT

Disraeli's government legally compels all municipal councils to provide clean water, sewers, and refuse collection, ending laissez-faire.

• LIVING TIMELINE SKETCHPAD & CAUSAL SYNTHESIS (1854-1900)

Connect Pasteur's swan-neck flask → Lister's carbolic spray → Bazalgette's brick sewer tunnels.

Pasteur opened the door; Lister, Koch, and Bazalgette marched through it.

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. Which ancient Greek physician created the Theory of the Four Humours?

2. Which Roman doctor introduced the Theory of Opposites?

3. In what year did Andreas Vesalius publish De Humani Corporis Fabrica?

4. What anatomical error of Galen did Vesalius disprove regarding the heart?

5. What did William Harvey prove about blood flow in 1628?

6. Which English doctor was nicknamed the 'English Hippocrates' for bedside observation?

7. What Latin term refers to poisonous, foul-smelling air blamed for epidemics?

8. In what year did the Great Plague strike London, killing 100,000+ people?

9. What Latin motto meaning 'Take nobody's word for it' was adopted by the Royal Society?

10. What was the primary function of medieval monastic hospitals?

• KEY VOCABULARY TASK

TERMINOLOGY

Define **Spontaneous Generation** and explain how Pasteur's 1861 **Germ Theory** scientifically disproved it:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which ideas about the cause of disease in the eighteenth century (c1700–c1800) were similar to ideas in the seventeenth century (c1500–c1700). [4 marks]

Hint: Focus on the universal continued belief in miasma (foul rotting air) causing epidemics like typhus and smallpox among both doctors and the public.

Stems: One way ideas about causes were similar was the continued belief in... • In the seventeenth century... • Similarly, in the eighteenth century...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which laboratory methods for investigating disease in the late nineteenth century (c1870–c1890) were different from the seventeenth century. [4 marks]

Hint: Contrast Koch's microscopic staining with aniline dyes, agar plates, and microphotography with Sydenham's bedside observation of visible outward symptoms.

Stems: One way methods differed was... • In the seventeenth century, doctors relied on... • In contrast, in the late nineteenth century, Koch used...

EXTENDED WRITING ASSESSMENT

Stimulus: Louis Pasteur's Germ Theory (1861) Robert Koch's work on bacteria (You must also use information of your own)

1. PASTEUR & GERM THEORY (1861)

2. KOCH'S BACTERIOLOGY (1876-83)

3. BRITISH ACCEPTANCE & MIASMA

ANALYTICAL CONNECTIVES:

KEY VOCABULARY BANK:

Task: Using the structure strip above, write 1-2 detailed analytical paragraphs below (continue in your exercise book for full timed paper):

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

☒ Done

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what year did Edward Jenner perform his famous cowpox test on James Phipps?

2. Who brought the dangerous practice of smallpox inoculation (variolation) to Britain in 1721?

3. What milder dairy disease gave milkmaids immunity from deadly smallpox?

4. How much money did Parliament grant Jenner in 1802 and 1807 to support vaccination?

5. In what year was infant smallpox vaccination made legally compulsory in Britain?

6. Name the protest organisation formed by citizens who opposed mandatory vaccination.

7. Why did the Royal Society initially refuse to publish Jenner's 1796 findings?

8. In what year did the British government formally ban dangerous smallpox inoculation?

9. Which French scientist discovered Germ Theory in 1861, explaining why vaccines worked?

10. What Latin word for 'cow' is the root of Jenner's term 'vaccination'?

• KEY VOCABULARY TASK

TERMINOLOGY

Explain the crucial medical distinction between **Smallpox Inoculation (Variolation)** and Jenner's **Cowpox Vaccination**:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which smallpox inoculation was similar to smallpox vaccination in eighteenth-century Britain. [4 marks]

Hint: Focus on the shared biological objective: deliberately introducing a mild dose of pox matter into the skin to stimulate natural bodily resistance.

Stems: One way inoculation and vaccination were similar was... • Inoculation involved... • Similarly, Jennerian vaccination...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which smallpox vaccination was different from smallpox inoculation in eighteenth-century Britain. [4 marks]

Hint: Contrast safety and contagion: cowpox vaccination was non-fatal and non-contagious, whereas smallpox inoculation risked killing the patient or starting a full epidemic.

Stems: One way vaccination differed was its superior safety... • Inoculation carried the risk of... • In contrast, cowpox vaccination...

Stimulus: Smallpox vaccination (1796) The Public Health Act (1875) *(You must also use information of your own)*

1. JENNER'S 1796 BREAKTHROUGH Explain Jenner's empirical milkmaid observation, inoculating James Phipps with cowpox (1796), and publishing conclusive case studies (1798).	2. STATE BACKING & COMPULSION Explain parliamentary grants (£30,000 in 1802/1807), banning variolation (1840), the 1852 Compulsory Vaccination Act, and 1871 registration fines.	3. OPPOSITION & CRITERIA EVALUATION Evaluate resistance (Anti-Vaccination League, Royal Society skepticism, inoculator profits) vs other prevention milestones (clean water and sewers under the 1875 Act).
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smallpox • variolation / inoculation • Lady Mary Wortley Montagu • Edward Jenner • cowpox • Sarah Nelmes • James Phipps • Royal Jennerian Society (1803) • parliamentary grants (£30,000) • 1852 Compulsory Vaccination Act • 1871 registration • Anti-Vaccination League

[illegible]☐ Done

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what war did Florence Nightingale serve as superintendent of female nursing from 1854 to 1856?

2. Name the British military hospital near Constantinople where Nightingale was stationed.

3. To what percentage did Nightingale's sanitary reforms reduce the mortality rate at Scutari?

4. What title was given to Florence Nightingale by the British press during the Crimean War?

5. What landmark book did Nightingale publish in 1859 setting down rules for nurse training?

6. Which London hospital housed the first professional Nightingale Training School for Nurses in 1860?

7. What architectural hospital design featured long, narrow wards with large windows for cross-ventilation?

8. What fatal bacterial infection was notoriously spread in pre-1850 unwashed hospital wards?

9. What Charles Dickens character typified the drunken, untrained, neglectful pre-reform nurse?

10. What was the primary function of medieval monastic hospitals in England?

• KEY VOCABULARY TASK

TERMINOLOGY

Define the **Pavilion Plan** in hospital architecture and explain how it eliminated cross-infection between wards:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which hospital care in the early eighteenth century (c1700–c1750) was similar to hospital care in the medieval period (c1250–c1500). [4 marks]

Hint: Focus on the shared reality that hospitals offered care, warmth, and palliative shelter rather than modern antiseptic cures, with high infection risks in shared beds.

Stems: One way hospital care was similar was... • In the medieval period... • Similarly, in the early eighteenth century, voluntary hospitals...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which hospital design in the late nineteenth century (c1860–c1890) was different from the early nineteenth century. [4 marks]

Hint: Contrast Nightingale's pavilion plan (separate isolation wings, cross-ventilation, glazed wipeable walls) with unventilated, cramped pre-1850 wards.

Stems: One way hospital design differed was... • In the early nineteenth century, wards were... • In contrast, under Nightingale's pavilion design...

EXTENDED WRITING ASSESSMENT

Stimulus:

Changes in hospital design and training (1859–60)

STRUCTURE STRIP • 3-PARAGRAPH ANALYTICAL PLAN

Explain how Nightingale and 38 nurses cleaned filthy barracks, unblocked sewers, provided fresh bedding, and cut Scutari mortality from ~40% to 2%.

Explain Notes on Hospitals (1863), the pavilion plan (cross-ventilation, separate infectious wings, wipeable glazed tiles), and halting hospital gangrene.

Explain Notes on Nursing (1859), the Nightingale School at St Thomas' (1860), transforming nursing from uneducated domestic work into a respected career.

A primary catalyst for hospital reform was... • In the Crimea, Nightingale demonstrated that... • Furthermore, this was codified through... • Consequently, civilian hospitals transformed from... • This directly resulted in...

Scutari military hospital • Crimean War (1854–56) • mortality rate (40% to 2%) • Notes on Nursing (1859) • Notes on Hospitals (1863) • pavilion plan • cross-ventilation • St Thomas' Hospital (1860) • professionalisation • sanitation

[illegible]

☒ Done

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. What Edinburgh professor of midwifery discovered the anaesthetic properties of chloroform in 1847?

2. Which early anaesthetic gas was discovered by Humphry Davy in 1799?

3. Which monarch famously used chloroform during the birth of Prince Leopold in 1853?

4. Why did death rates from infection increase during the 'Black Period' of surgery (1847–1865)?

5. Which Scottish surgeon pioneered antiseptic surgery using carbolic acid in 1865?

6. Whose 1861 scientific discovery directly inspired Lister to investigate wound sepsis?

7. To what percentage did Lister's carbolic acid treatments reduce amputee mortality in Glasgow?

8. What steam-pressure sterilisation machine was invented by Charles Chamberland in 1881?

9. What surgical innovation did William Halsted introduce at Johns Hopkins Hospital in 1890?

10. Which 16th-century French barber-surgeon famously replaced boiling oil with soothing rose-oil lotion?

• KEY VOCABULARY TASK

TERMINOLOGY

Explain the fundamental operational difference between **Antiseptic Surgery** (Lister) and **Aseptic Surgery** (c.1890):

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which surgical conditions during the 'Black Period' (1847–1865) were similar to Renaissance surgery. [4 marks]

Hint: Focus on the persistence of dirty operating coats, unwashed instruments, and total lack of hygiene leading to fatal gangrene and sepsis.

Stems: One way surgery was similar was the uncontrolled risk of infection... • In the Renaissance period... • Similarly, in the 1850s, surgeons still operated in...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which surgery in 1890 was different from surgery in 1840 in Britain. [4 marks]

Hint: Contrast complete patient anaesthesia (chloroform) and germ exclusion (aseptic surgery, autoclaves, rubber gloves) with conscious patients held down in agony and sepsis.

Stems: One way surgery in 1890 differed was the conquest of both pain and infection... • In 1840, patients were... • In contrast, by 1890, surgery was...

EXTENDED WRITING ASSESSMENT

(You must also use information of your own)

Explain the transition from killing germs with harsh chemicals (antiseptic) to excluding germs entirely: autoclaves (1881), boiled rubber gloves, and surgical scrubs.

laughing gas • ether • James Young Simpson • chloroform (1847) • Queen Victoria (1853) • "Black Period" of surgery • sepsis • gangrene • Joseph Lister • carbolic acid spray (1865) • antiseptic surgery • aseptic surgery • autoclave (1881)

[illegible]

11/14

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what year did the first deadly epidemic of Asiatic cholera arrive in Britain?

2. What government economic philosophy meaning 'leave alone' opposed sanitary spending?

3. Whose landmark 1842 report proved poor sanitary conditions caused disease and poverty?

4. Why was the 1848 First Public Health Act largely ineffective across Britain?

5. Which London doctor mapped the 1854 Soho cholera outbreak to the Broad Street water pump?

6. What physical action halted the cholera epidemic in Soho in September 1854?

7. What environmental crisis in hot summer 1858 forced Parliament to fund London's sewers?

8. Which civil engineer designed and built London's revolutionary 82-mile brick sewer network?

9. Which political party leader passed the compulsory 1875 Second Public Health Act?

10. Name one compulsory requirement placed on local town councils by the 1875 Public Health Act.

• KEY VOCABULARY TASK

TERMINOLOGY

Explain the fundamental difference between the **1848 Public Health Act (Permissive)** and the **1875 Public Health Act (Compulsory)**:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which civic reactions to the 1848 cholera epidemic were similar to reactions during the Great Plague of 1665. [4 marks]

Hint: Focus on the continued belief in miasma leading authorities to burn tar barrels, wash streets with vinegar, and rely on local quarantine without tackling water contamination.

Stems: One way civic reactions were similar was the reliance on miasma theories... • In 1665, watchmen... • Similarly, in 1848, local officials...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which government attitudes to public health in 1875 were different from attitudes in 1800. [4 marks]

Hint: Contrast the compulsory state intervention of the 1875 Act (enforcing sewers, clean water, housing standards) with the total laissez-faire inaction of 1800.

Stems: One way attitudes differed was the total rejection of laissez-faire... • In 1800, the British government believed... • In contrast, by 1875, Parliament passed legislation that compelled...

Stimulus:

The Public Health Act (1875)

STRUCTURE STRIP • 3-PARAGRAPH ANALYTICAL PLAN

Evaluate the end of laissez-faire through compulsory national regulation (clean water, sewage, housing checks, food purity) vs Snow's localized Soho breakthrough.

laissez-faire • miasma theory • cholera epidemics (1831, 1848, 1853–54) • John Snow • Broad Street pump (1854) • Soho spot map • cesspools • Edwin Chadwick (1842) • 1848 Public Health Act (permissive) • Great Stink (1858) • Joseph Bazalgette • 1875 Public Health Act (compulsory)

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☒ Done

PUPIL NAME: _____ TARGET: _____ PREDICTED: _____

Date Completed	Lesson • Specification Focus	Q3 Practice [4m]	Extended Response	Lesson Total
	KT3.1: Pasteur & Koch	Q3 Diff: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT3.2: Edward Jenner	Q3 Diff: [/ 4]	Q5/6 Essay: [/ 20]	[/ 24]
	KT3.3: Florence Nightingale	Q3 Sim: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT3.4: Simpson & Lister	Q3 Diff: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT3.5: John Snow & 1875 Act	Q3 Diff: [/ 4]	Q5/6 Essay: [/ 20]	[/ 24]
CUMULATIVE ASSESSMENT TOTALS		Q3 Total: [/ 20]	Extended: [/ 76]	[/ 96]

TEACHER FORMATIVE ASSESSMENT • FEEDBACK

Effort: [1 • 2 • 3 • 4 • 5]

WHAT WENT WELL (WWW):

EVEN BETTER IF (EBI):

Teacher Signature: _____

Date: _____

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE

ONLINE RECALL

KT3.1

Pasteur & Koch



SCAN TO QUIZ

Best Score: [/ 8]

KT3.2

Edward Jenner



SCAN TO QUIZ

Best Score: [/ 8]

KT3.3

F. Nightingale



SCAN TO QUIZ

Best Score: [/ 8]

KT3.4

Simpson & Lister



SCAN TO QUIZ

Best Score: [/ 8]

KT3.5

Snow & 1875 Act



SCAN TO QUIZ

Best Score: [/ 8]